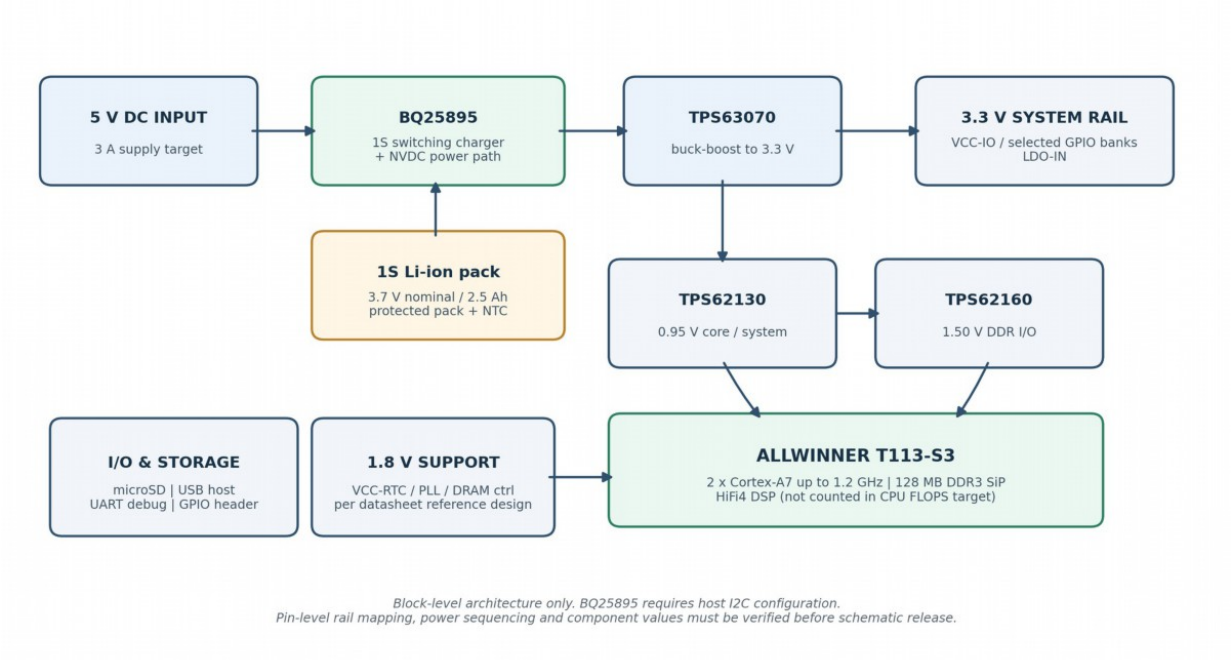
OPENPOWER COMPUTE-01

A battery-backed, sub-5-watt embedded computing board

Design target: 3-4 GFLOP/s FP32 vector/matrix workload

Design Research Paper

Revision 0.9 | 08 November 2026



Block-level architecture only. BQ25895 requires host I2C configuration.
Pin-level rail mapping, power sequencing and component values must be verified before schematic release.

Design status: PRE-SCHEMATIC PROPOSAL

Not yet a pin-complete schematic, validated prototype, frozen BOM or manufacturing-ready Gerber package.

Abstract

This paper proposes OpenPower Compute-01, a compact, battery-backed embedded computer intended to demonstrate that useful general-purpose numerical computing can be delivered within a five-watt board-level power envelope. The proposed processor is the Allwinner T113-S3 system-on-chip (SoC), which integrates two Arm Cortex-A7 CPU cores, a HiFi4 DSP and 128 MB of DDR3 memory in a system-in-package implementation. The integrated memory removes the need to route an external DDR bus, reducing the complexity of a first open hardware design. The board is designed around a single-cell 3.7 V lithium-ion battery and a charger with system load-sharing, followed by regulated 3.3 V, 1.5 V, 1.8 V and approximately 0.95 V power domains. The paper evaluates a 3-4 GFLOP/s FP32 target using a published Cortex-A7 GFLOPS reference figure and a transparent frequency/core-count extrapolation. That value is a design target, not a vendor guarantee or a measurement from a completed T113-S3 board. The paper defines a minimal embedded-Linux use case, candidate components, rail sequencing constraints, power and runtime calculations, PCB layout guidance, and a verification protocol. The outcome is a defensible Rev-A architecture ready for detailed electrical questions and schematic capture, but not yet ready for fabrication.

Keywords: low-power computing; embedded Linux; open hardware; Allwinner T113-S3; Cortex-A7; NEON; GFLOPS per watt; battery power path; PCB design; energy efficiency.

Central research proposition A dual-core Cortex-A7 board may reach a measured 3-4 GFLOP/s on carefully optimized FP32 vector/matrix workloads while drawing no more than 5 W from the battery terminals during a battery-only benchmark. This must be demonstrated by simultaneous benchmark and power measurements.

1. Introduction and problem statement

Many embedded boards emphasize connectivity, displays, cameras or large software stacks. OpenPower Compute-01 instead prioritizes a deliberately narrow experiment: how much useful floating-point work can a commercially documented, low-cost application processor deliver per watt when unnecessary peripherals are omitted? The board is not intended to compete with a laptop or desktop. It is intended to run a small Linux userland, execute several lightweight applications, and provide an openly documented platform for measuring performance and energy use.

The key engineering metric is not raw peak FLOPS alone. It is measured workload performance divided by measured DC input power. A board that reports a high arithmetic peak but consumes excessive power, throttles thermally, or cannot sustain its software workload would fail the project objective. Accordingly, the design treats the 3-4 GFLOP/s range as a workload-specific acceptance target and the 5 W figure as a whole-board input ceiling.

The phrase “a few MB of RAM” requires a practical correction. A Linux-capable application processor board with a few user applications needs materially more than 2-8 MB. The selected T113-S3 integrates 128 MB of DDR3. This remains modest by modern PC standards, but it is a realistic starting point for a stripped-down embedded Linux system. A design capped at only a few megabytes would need an MCU-class architecture and would not realistically meet the same 3-4 GFLOP/s general-purpose computing goal.

2. Design requirements and acceptance criteria

Requirement	Rev-A proposal	Verification
Compute target	3.0-4.0 GFLOP/s FP32 on an optimized, documented two-core NEON matrix/vector benchmark.	Measured benchmark; publish compiler flags, clock, data size and median result.

Requirement	Rev-A proposal	Verification
Power ceiling	No more than 5.0 W drawn from the 1S pack during battery-only compute test.	Measure pack-terminal voltage and current simultaneously with benchmark; repeat adapter-mode test with charge disabled.
Processor / memory	Allwinner T113-S3; dual Cortex-A7 up to 1.2 GHz; 128 MB integrated DDR3.	Confirm exact silicon marking, clock configuration and memory availability.
Battery	Single-cell protected Li-ion/Li-poly pack, 3.7 V nominal, 2,500 mAh target, NTC.	Battery-protection, charge-temperature and switchover tests.
Storage	microSD boot/root file system; optional SPI-NOR footprint deferred.	Cold boot, reboot, read/write and recovery tests.
Applications	Small Linux shell, math benchmark, calculator, sensor logger and lightweight command-line utilities.	Run representative applications concurrently within memory budget.
I/O	One USB host, USB device/serial or debug access, UART header, reset/boot buttons and a small GPIO/I2C/SPI expansion header.	Interface enumeration and loopback tests.
Board complexity	Four-layer FR-4 PCB; no external DDR3; no display, Wi-Fi or Ethernet in Rev A.	Layout review, signal-integrity checks and assembly inspection.

Out of scope for Rev A: a graphical desktop, browser-heavy applications, 4K video, wireless networking, AI inference acceleration, and any claim that the board is a replacement for a general-purpose PC. Optional display and radio modules can be explored after the base design passes the power and performance tests.

3. Processor selection and component availability

3.1 Why the T113-S3 is selected

The T113-S3 is a practical compromise between a very small single-core processor and a modern, higher-cost application processor. The Allwinner data sheet specifies two Cortex-A7 cores, a HiFi4 DSP, 128 MB of integrated DDR3, USB 2.0 device/host interfaces, SD/MMC host interfaces and a 128-pin eLQFP package measuring 14 mm by 14 mm. The product listing is available from LCSC under part number C5197687. The exact distributor stock, price and orderable suffix must be checked again before procurement because availability is dynamic [1, 2, 7].

The MYIR T113-S3 system-on-module specification independently lists two Cortex-A7 cores running up to 1.2 GHz and 128 MB of integrated DDR3, and documents a Linux 5.4 software path. It also demonstrates that commercial module designers use the T113-S3 in embedded Linux applications [2]. The open-source TESS project provides a further example of a compact T113-S3 board architecture with integrated memory and battery-related circuitry, although its design is a reference for study rather than a drop-in schematic for this board [8].

3.2 Comparison with the smaller V3s option

Candidate	CPU / RAM	Strength	Why not selected
Allwinner V3s	1 x Cortex-A7 up to 1.2 GHz; 64 MB SiP DDR2.	Very simple, small and low-power embedded Linux platform.	One A7 core does not provide a credible path to the 3-4 GFLOP/s target.
Allwinner T113-S3	2 x Cortex-A7 up to 1.2 GHz; 128 MB SiP DDR3.	Two CPU cores, integrated RAM, LQFP package and usable embedded Linux ecosystem.	More power rails and sequencing complexity than an MCU or V3s.

The V3s remains a valid alternative for a lower-performance ultra-low-power edition, but the requested arithmetic target makes the dual-core T113-S3 the more defensible first choice. Integrated DDR is the important manufacturability advantage: the board does not need a separate external DDR3 chip and its associated high-speed address/data routing.

3.3 Candidate component set

Function	Candidate / specification	Engineering status
Application processor	Allwinner T113-S3, eLQFP-128, LCSC C5197687.	Selected; check live stock, package drawing and official pin-mux data before layout.
Battery charger and power path	Texas Instruments BQ25895, single-cell switching charger, NVDC power path, 5 A maximum charge rating and up to 3.25 A input-current limit.	Preferred candidate; choose a 4.20 V charge configuration, verify I2C defaults, inductor, thermals and pack settings.
3.3 V battery-to-system regulator	TI TPS63070 buck-boost, 2.0-16 V input, 2.5-9 V output, 3.6 A switch-current limit.	Candidate; calculate inductor, output-current margin and thermal performance for this load.
0.95 V and 1.50 V regulators	TI TPS62130 synchronous buck, up to 3 A, for the 0.95 V core/system rail; TI TPS62160, up to 1 A, for the 1.50 V DDR rail.	Separate rails; verify load transients, soft-start, stability and thermals against the exact T113 reference circuit.
Storage	microSD socket and removable microSD card.	Selected for simplicity; boot-device strap and supported boot mode must be checked in the T113 user manual.
Clocking	24 MHz reference crystal; 32.768 kHz RTC crystal optional.	Values and load capacitors must follow the processor reference circuit.
Input protection / USB power	Fuse or resettable protector, reverse/over-voltage protection and USB load switch where required.	Exact parts remain to be chosen after connector and adapter requirements are fixed.

The listed parts are candidate engineering selections, not a locked purchasing BOM. The BQ25895, TPS63070, TPS62160 and TPS62130 have published manufacturer specifications supporting the proposed roles [4-6, 12]. The final exact package suffix, footprints, external component values and live distributor availability will be confirmed before schematic and Gerber release.

4. Performance analysis: a defensible 3-4 GFLOP/s target

4.1 Published benchmark basis

The open-source MegPeak repository lists 1.902002 GFLOP/s for a Cortex-A7 at 1.0 GHz. Its README describes peak-computation and instruction-throughput testing, but does not fully identify the test platform, arithmetic precision, compiler flags or exact instruction mix behind that table entry. It is therefore an indicative reference point, not a verified FP32 result for the T113-S3 [3]. MYIR specifies the T113-S3 Cortex-A7 cores at up to 1.2 GHz [2].

4.2 Scaling model and interpretation

A first-order estimate scales the published single-core result linearly with clock frequency and then with the number of active cores:

Performance model 1.902 GFLOP/s x (1.2 GHz / 1.0 GHz) x 2 cores = 4.565 GFLOP/s (calculated extrapolation, not measured silicon or board performance).

A 3.0-4.0 GFLOP/s measured result would therefore require approximately 66-88% of this simple scaled peak estimate. That is ambitious but plausible for a highly optimized FP32 NEON matrix-multiplication kernel with both cores active. It is not a safe expectation for arbitrary application code: memory access patterns, compiler output, OS scheduling, cache misses, synchronization overhead, temperature and frequency limits can all reduce throughput. The T113-S3 includes a HiFi4 DSP, but the DSP is deliberately excluded from this CPU-only target because its toolchain and workload mapping have not been validated in this proposal.

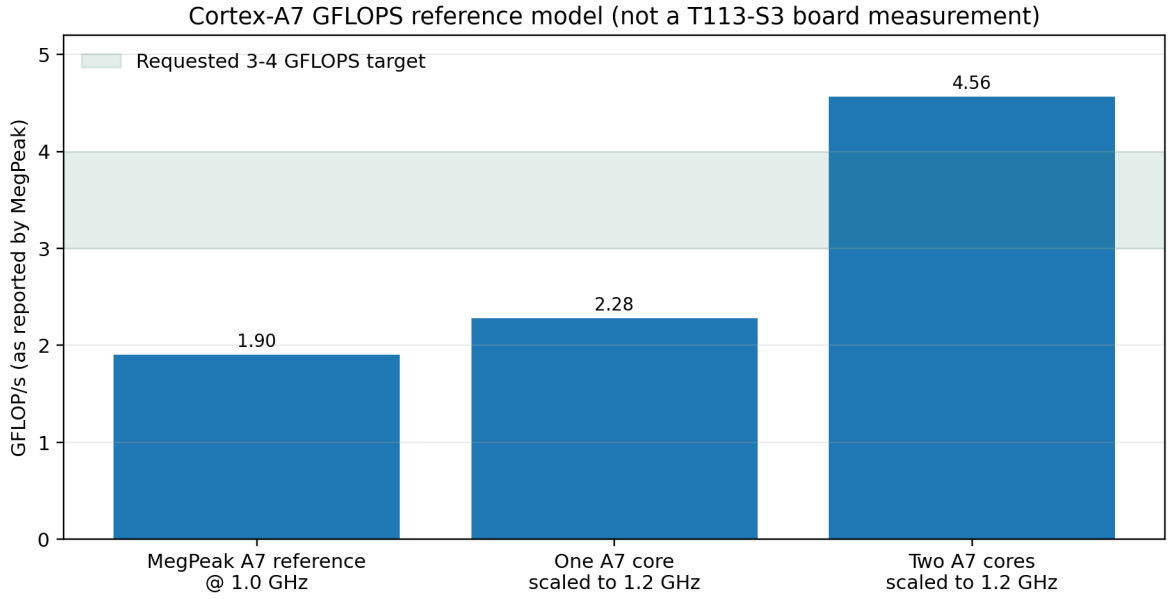


Figure 1. MegPeak-published Cortex-A7 GFLOPS reference and simple frequency/core-count scaling. The benchmark README does not specify precision or full test conditions; the scaled bars are calculations, not measured T113-S3 results.

4.3 Benchmark definition

The board should publish two different numbers rather than conflating peak and sustained performance: (a) MegPeak instruction/peak-compute result per core and across cores, and (b) a sustained FP32 SGEMM or matrix-vector result. For a square N x N matrix multiplication, the conventional operation count is approximately 2N^3 floating-point operations; throughput is that operation count divided by elapsed seconds. The report must identify whether fused multiply-add is counted as two FLOPs, record the matrix sizes and library/compiler, pin the CPU frequency, and report the median of at least five runs.

- Use an ARMv7 NEON-capable toolchain and document the exact compiler flags; do not silently use host-machine results.
- Run a single-core test and then a two-core test, pinning threads to separate CPU cores where possible.
- Run at least one small working-set test and one larger test that exercises the memory subsystem.
- Measure DC input voltage and current at the same time as the benchmark. Report average, peak and settled power separately.
- Repeat the test after a 15-minute thermal soak and record CPU frequency and temperature to identify throttling.

The acceptance target is reached only if the measured two-core FP32 result is at least 3.0 GFLOP/s while battery-terminal input power remains below 5.0 W during the same battery-only run. A 4.0 GFLOP/s result is the stretch goal. If MegPeak reports 3-4 GFLOP/s but sustained SGEMM does not, both results must be published transparently.

5. Electrical architecture

5.1 Power-path concept

The proposed battery architecture uses a protected single-cell Li-ion/Li-polymer pack and the Texas Instruments BQ25895 switching charger with narrow-voltage DC (NVDC) power-path management. The device supports a 3.9-14 V input, a 5 A maximum switch-mode charge path, up to 3.25 A programmable input-current limit, battery-temperature monitoring and an 11 mOhm BATFET specified for battery discharge up to 9 A [4]. In battery-only operation, the BATFET connects the cell to SYS; with an input source present, the charger manages the system rail and battery charging, and the battery can supplement the input when required. This higher-current switching architecture is a better candidate for the approximately 2 A low-cell-voltage current implied by a 5 W load than the earlier MCP73871 linear-charger option. Microchip's MCP73871 design guide gives a typical 1.65 A system load and 1.5 A minimum from a wall adapter, making it a less comfortable choice for this target [11]. The battery pack must still include independent cell protection; the charger IC is not a substitute for a protected pack.

A 5 V / 3 A regulated DC input is recommended for Rev A, using a locking or clearly polarized connector. This provides headroom for the board load and controlled battery charging; the BQ25895 supports 3.9-14 V input and up to 3.25 A input-current limit [4]. USB-C may be used only after its current/PD negotiation and protection implementation is explicitly designed. The BQ25895 SYS node feeds a TPS63070 buck-boost regulator configured for 3.3 V. The TPS63070's 2.0-16 V input range accommodates the BQ25895 system rail when an adapter is present and the battery voltage when operating from the cell [5]. The BQ25895 is I2C-configurable, so the T113-S3 must initialise or verify safe charger settings early in boot; hardware defaults, charge enable and input-current limits must be checked before the schematic is frozen.

5.2 Processor supply rails

T113-S3 domain	Data-sheet nominal voltage	Proposed supply method / design note
VDD-CORE0/1 and VDD-SYS0/1/2	0.95 V typical (recommended operating range 0.85-0.99 V).	Dedicated synchronous buck from 3.3 V. Candidate TPS62130 set to 0.95 V; keep rail copper short and well decoupled.
VCC-DRAM0/1	1.50 V typical (1.425-1.575 V).	Separate 1.50 V buck rail. Candidate TPS62160; do not share the 0.95 V rail.
VDD18-DRAM, VCC-PLL, VCC-RTC and 1.8 V domains	1.80 V typical.	Use the SoC internal LDO output(s) only as specified by the vendor reference circuit, or add an external 1.8 V LDO if required by the pin-level design.
VCC-IO and selected VCC-PD/PE/PG banks	3.30 V for the selected 3.3 V I/O configuration.	Main 3.3 V system rail. Confirm each I/O bank voltage against its pin mux and attached devices.
LDO-IN	3.30 V typical.	Feed from the regulated 3.3 V system rail as required by the data sheet.
LDOA-OUT / LDOB-OUT	LDOA nominal 1.8 V; LDOB default 1.35 V with listed options including 1.50 V and 1.80 V.	These are on-chip regulator outputs. Do not drive them as if they were external input rails; connect and load them only as the data sheet/reference schematic directs.

The current data sheet gives 0.95 V as the typical recommended CPU/system rail, while its power-sequence illustration labels the rail 0.9 V. Since 0.9 V is within the stated 0.85-0.99 V operating interval, the values are not electrically contradictory, but the Rev-A nominal design value should be 0.95 V unless a T113-S3 reference schematic or vendor FAE confirms a different configuration. The final schematic must follow the exact data-sheet revision and official reference circuit [1].

5.3 Power sequencing and reset

The Allwinner data sheet specifies that VCC-RTC must ramp no later than the other power rails; VCC-IO must ramp before VDD-SYS and VDD-CORE with a minimum delay of 2 ms; VCC-DRAM must be stable before SDRAM initialization; reset must remain low until the rails have stabilized for more than 64 ms; and the 24 MHz clock begins oscillating after reset is released [1]. This sequence is a first-order design constraint, not an optional firmware detail.

- Generate a power-good / reset solution that guarantees the required rail order and reset hold time across adapter and battery startup.
- Place local bypass capacitors at every power pin group using the data sheet and reference schematic values.
- Validate the rail ramp waveforms with an oscilloscope on the first assembled board before Linux boot testing.
- Keep unused video, audio and interface domains in the correct powered or disabled state defined by the reference schematic; do not leave required power pins floating.

6. Battery sizing, input power and energy efficiency

6.1 Battery pack

The proposed battery is a protected 1S Li-ion or Li-polymer pack rated 3.7 V nominal and 2,500 mAh, equivalent to approximately 9.25 Wh nominal stored energy. Select a pack with integrated over-current and over-discharge protection and an NTC temperature lead. Configure the BQ25895 initially for a conservative 0.5-1.0 A charge current; do not use its 5 A silicon capability as the pack charge setting. Increase the current only if the cell data sheet, connector, copper and thermal tests support it. The battery should be rated for at least 2.5 A continuous discharge, with cell-specific derating and protection limits confirmed before purchase.

6.2 Battery current and run-time calculation

At a 5.0 W regulated-rail load, 3.0 V low battery voltage and an assumed 85% aggregate conversion efficiency, the battery current estimate is:

Battery-current estimate I = P / (V x efficiency) = 5.0 W / (3.0 V x 0.85) = 1.96 A. Specify a protected pack capable of at least 2.5 A continuous discharge, subject to the battery maker data sheet.

The ideal energy-based run time at a 5 W rail load is 9.25 Wh / 5 W = 1.85 hours before conversion loss, cutoff reserve, battery aging and temperature effects. Applying the 85% planning efficiency gives approximately 1.57 hours. This is an estimate, not a measured run time. Actual run time will depend on the average workload power and the battery discharge curve.

6.3 Whole-board power budget

The five-watt figure is a maximum system-power budget for the complete board, not an assertion that the CPU alone consumes five watts. For an unambiguous acceptance test, run from the battery only and measure voltage and current directly at the battery connector; this captures all board power and avoids under-counting if the charger path lets the battery supplement a weak adapter. The T113-S3 data sheet provides maximum current ratings for its power terminals and states that the listed ratings are worst-case estimates and actual application estimates may be lower [1]. Multiplying those current ratings by the selected nominal rail voltages produces a conservative rail-sizing envelope of approximately 3.14 W if every listed domain were simultaneously at its maximum rating. That arithmetic is useful for initial rail sizing but must not be

represented as measured typical SoC consumption; some domains are unused in this minimal design and maximum values may not coincide.

At an assumed 85% conversion efficiency, 3.14 W of rail load would require approximately 3.70 W from the battery before adding removable storage, USB peripherals and any optional display or wireless module. This leaves limited headroom under a 5 W ceiling. The design therefore excludes display, Wi-Fi, Ethernet and high-current USB accessories from Rev A. The actual acceptance test must directly measure the finished board rather than relying on the rail-envelope estimate.

Metric	Value	Status
Requested compute throughput	3.0-4.0 GFLOP/s	Measured acceptance target, not guaranteed.
Board power ceiling	5.0 W at battery terminals	Design constraint to be verified under battery-only compute load.
Efficiency at 3 GFLOP/s and 5 W	0.60 GFLOP/s/W	Lower-end target-derived ratio.
Efficiency at 4 GFLOP/s and 5 W	0.80 GFLOP/s/W	Stretch-target ratio at the ceiling.
Energy per operation at 5 W	1.67 nJ/FLOP at 3 GFLOP/s; 1.25 nJ/FLOP at 4 GFLOP/s.	Calculated from the requested target and battery-side power ceiling; not measured.

7. PCB implementation strategy

A four-layer FR-4 PCB is recommended for the first layout. A two-layer board is not recommended for a dual-core application processor with several supply domains and USB/SD interfaces because it makes ground return paths, decoupling and power distribution unnecessarily difficult. The integrated DDR3 eliminates external DDR routing, but the processor still has dense fine-pitch fanout, high-speed USB and SD signals, clocking and multiple regulated rails.

Layer	Suggested role
L1 / Top	T113-S3, charger, regulators, decoupling, crystal and short critical signals.
L2 / Inner 1	Continuous ground reference plane; avoid splits under USB and clock routes.
L3 / Inner 2	Power distribution and low-speed routing; keep switching-current loops compact.
L4 / Bottom	Secondary signal routing, connectors, microSD and user I/O.

A provisional board envelope around 50 mm x 50 mm is reasonable for discussion, but it is not frozen. The final outline depends on the chosen 5 V input connector, battery connector, USB host receptacle, microSD orientation, mounting holes and desired GPIO header. The design should place the high-current power path near the input and battery connector, the SoC near the center, its decoupling capacitors tightly around the power pins, and the high-speed connectors so that their routes remain short.

- Keep the buck-boost and buck converter switch nodes, inductors and input/output capacitors in compact loops. Do not route crystal or USB/SD signals through switching-node regions.
- Provide a continuous ground reference and multiple ground vias near the SoC decoupling capacitors and connector shields.
- Use controlled-impedance routing for USB 2.0 differential pairs according to the PCB manufacturer's stack-up; avoid unnecessary stubs and vias.
- Keep the 24 MHz crystal and its load components adjacent to the processor clock pins, following the reference layout.
- Include clearly labelled test pads for 3.3 V, 1.8 V, 1.5 V, 0.95 V, SYS, battery and ground.

- Use a thermal copper area around the SoC and converters. Validate temperature under a sustained compute load; add a small heatsink footprint only if measurements require it.

Rev-A I/O recommendation 5 V input, one USB 2.0 host port, microSD slot, 3-pin UART debug header, reset and boot buttons, status LEDs, and a modest 2.54 mm GPIO/I2C/SPI expansion header. No display, Ethernet or Wi-Fi on the first revision.

8. Firmware and software plan

The initial software image should be a small embedded Linux distribution built with Buildroot or the T113-S3 board support package. The MYIR platform lists Linux 5.4 support, and existing T113-S3 boards demonstrate the SoC's use as an embedded Linux processor [2, 8]. Kernel and bootloader compatibility must be pinned to the exact board design and chosen boot media; do not assume every upstream Linux version supports every board without device-tree and driver work.

Application	Purpose	Memory strategy
BusyBox shell and basic utilities	Interactive terminal, file operations and system management.	Keep the base root filesystem small.
FP32 compute test	NEON matrix multiplication and benchmark reporting.	Use fixed, documented matrix sizes; avoid large allocation.
Command-line calculator	Demonstrate ordinary arithmetic and numeric routines.	Native C/C++ executable.
Sensor / GPIO logger	Record I2C/SPI/UART sensor data to microSD.	Event-driven, bounded buffers.
Optional Python 3	Small scripts and educational use.	Include only if the final rootfs and memory budget allow it.

The first version should be headless and managed through the UART debug console or USB serial. This is intentional: an LCD controller, touch panel and graphical stack would increase board cost, routing effort and power draw before the central compute-per-watt experiment has been demonstrated. A display can be added as a later carrier or expansion design.

9. Verification and validation plan

Test ID	Procedure	Pass condition / output
PWR-01	Measure all rails at startup, adapter present and battery-only.	Every rail within data-sheet tolerance; correct rail order; reset released only after required delay.
BOOT-01	Cold boot and warm reboot from microSD.	Repeatable boot, serial console available, no SD errors.
BAT-01	Run system from adapter, remove adapter, restore adapter.	No reset or data corruption during transition; charger behaves within datasheet limits.
CHG-01	Charge at conservative setting with representative system load.	Cell temperature remains in battery specification; input current remains within programmed limits.
CPU-01	Run MegPeak per core and with both cores.	Record clock, per-core and aggregate result, compiler and tool version.
GEMM-01	Run FP32 SGEMM with one and two worker threads.	Target 3.0-4.0 GFLOP/s on two cores; report matrix dimension and median runtime.

Test ID	Procedure	Pass condition / output
PWR-02	In battery-only mode, measure pack-terminal voltage/current simultaneously with CPU-01 and GEMM-01. Repeat a separate adapter-mode test with battery charge disabled.	Battery-side system power stays below 5.0 W for the qualifying benchmark run; adapter-mode result is separately reported.
THERM-01	Repeat qualifying test for at least 15 minutes.	No thermal shutdown; sustained clock and throughput reported; temperatures documented.
APP-01	Run shell, math demo and logger concurrently.	No out-of-memory event, reset or storage corruption.

The public release should include raw measurement logs, board revision, bill of materials with manufacturer part numbers, test environment, source code and reproducible build instructions. Results should distinguish measured values, calculated estimates and vendor maximum ratings.

10. Engineering risks and limitations

- Performance uncertainty: the 4.565 GFLOP/s figure is frequency/core-count scaling from a public Cortex-A7 benchmark; memory, compiler and silicon differences can make the actual T113-S3 result lower.
- Power uncertainty: no trustworthy board-level T113-S3 idle/full-load power measurement was identified in the reviewed documentation. The five-watt target therefore remains a verification requirement.
- Power sequencing: the SoC has multiple rails and a specified startup order. A design that powers the rails in the wrong order may fail to boot or may damage reliability.
- Battery safety: use a protected pack with an NTC lead. Do not connect an unprotected bare lithium cell or bypass cell protection. Battery charger settings must follow the actual cell manufacturer's specification.
- Component procurement: current availability and package suffixes change. Confirm all exact order codes, lifecycle status and footprints before ordering a PCB.
- RAM limit: 128 MB supports a small embedded system but not a full modern desktop environment with multiple browser tabs and heavy GUI applications.
- Open-source documentation: Allwinner's published data sheet and vendor reference circuits must be checked against the precise silicon revision; the design must not infer unlisted pin functions or power domains.

11. Conclusion

The Allwinner T113-S3 is a credible starting point for a modest open-source Linux computer intended to explore computing under a strict five-watt input budget. Its dual Cortex-A7 cores, 128 MB in-package DDR3 and eLQFP128 package provide a practical balance between computational capability and PCB accessibility. A published Cortex-A7 GFLOPS reference figure produces a calculated 4.565 GFLOP/s two-core, 1.2 GHz scaling estimate; the original test precision and detailed conditions are not fully documented. This supports setting a 3-4 GFLOP/s optimized FP32 target, but only a real prototype benchmark can establish whether that target is achieved.

The proposed single-cell battery system uses an actual load-sharing charger and conventional buck/buck-boost regulators. The processor's multiple supply rails and startup sequence are explicitly included in the architecture. A minimal four-layer board with microSD, UART, one USB host and a small GPIO expansion interface is recommended for Rev A. Display, wireless and Ethernet features should be deferred so that the first prototype can focus on measured compute throughput, power, thermal stability and battery failover.

Decision Proceed to detailed schematic capture only after confirming the selected T113-S3 package and official reference schematic, exact power-rail implementation, battery pack, connectors and physical constraints. Do not order fabrication from this block-level paper alone.

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Appendix A. Information required before schematic and Gerber generation

The following decisions are intentionally open. They determine the connector footprints, board outline, mounting geometry, battery routing, mechanical clearance and final power configuration. No Gerber file will be released until these are tested and the complete schematic passes ERC/DRC and independent review.

1. Physical dimensions and shape: square, rectangular, or a defined maximum size; mounting-hole count and spacing.
2. Power input: 5 V barrel jack, USB-C with CC current detection, or another connector; exact adapter rating.
3. Battery format: protected pouch pack or 18650 holder; desired capacity; pack connector and NTC pinout.
4. User I/O: USB-A host count, USB-C data/device port, microSD position, UART header type, GPIO connector and button/LED placement.
5. Display: none for Rev A, or a specific SPI/RGB/LVDS panel and connector.
6. Storage: microSD-only, or microSD plus onboard SPI NOR/eMMC if an exact source and boot plan are confirmed.
7. Assembly constraints: preferred PCB manufacturer, four-layer stack-up, copper weight, minimum trace/space and whether assembly will be factory SMT or hand-soldered.
8. Procurement region and preferred distributors so that each component can be checked for live stock and alternate parts.